

Task 5:

Valgrind Evidence:

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riets@LAPTOP-RAGPJAP3:/mnt/c/Users/botha/Downloads/COS-214-Practical-4-main/COS-214-Practical-4-main$ docker run --rm -it -v "$(pwd):/app" -w /app taskforge-env bash
root@7b971fc2446d:/app# valgrind --leak-check=full --show-leak-kinds=all ./taskforge
==9== Memcheck, a memory error detector
==9== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==9== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==9== Command: ./taskforge
==9==

=== Scenario 1: coordinated power-plant fire response ===
The city manager dispatches a nested incident hierarchy.
New emergency: Tshwane emergency command
New emergency: Central district response
New emergency: Industrial sector response
New emergency: Power-plant fire incident
  INDIVIDUAL VEHICLE SENT OUT FOR Fire engine 1 to the turbine hall
  PRIORITY P1 - trapped workers: dispatch first for Rescue ambulance to the turbine hall
  INDIVIDUAL VEHICLE SENT OUT FOR Rescue ambulance to the turbine hall
Completed Power-plant fire incident
Completed Industrial sector response
Completed Central district response
Completed Tshwane emergency command

Depth-first incident view:
- Central district response
- Industrial sector response
- Power-plant fire incident
- Fire engine 1 to the turbine hall
- Rescue ambulance to the turbine hall [priority: P1 - trapped workers] [audited]

Reverse incident view:
- Central district response

Audit trail for the stacked dispatch decorators:
1. Dispatch started: Rescue ambulance to the turbine hall [priority: P1 - trapped workers]
2. Dispatch completed: Rescue ambulance to the turbine hall [priority: P1 - trapped workers]

=== Scenario 2: new hazard during an active incident ===
A depth-first snapshot is created before a hazmat team is added.
Existing snapshot sees the new hazmat unit: no
Fresh snapshot sees the new hazmat unit: yes

Updated depth-first incident view:
- Fire engine 1 to the turbine hall
- Rescue ambulance to the turbine hall [priority: P1 - trapped workers] [audited]
- Hazmat unit to chemical storage [priority: P1 - escalating chemical risk] [audited]

Dispatching the updated power-plant incident:
New emergency: Power-plant fire incident
  INDIVIDUAL VEHICLE SENT OUT FOR Fire engine 1 to the turbine hall
  PRIORITY P1 - trapped workers: dispatch first for Rescue ambulance to the turbine hall
  INDIVIDUAL VEHICLE SENT OUT FOR Rescue ambulance to the turbine hall

  PRIORITY P1 - escalating chemical risk: dispatch first for Hazmat unit to chemical storage
  INDIVIDUAL VEHICLE SENT OUT FOR Hazmat unit to chemical storage
Completed Power-plant fire incident

TaskForge emergency response scenarios completed successfully.
==9==
==9== HEAP SUMMARY:
==9==   in use at exit: 0 bytes in 0 blocks
==9==   total heap usage: 140 allocs, 140 frees, 81,755 bytes allocated
==9==
==9== All heap blocks were freed -- no leaks are possible
==9==
==9== For lists of detected and suppressed errors, rerun with: -s
==9== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
root@7b971fc2446d:/app#
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